

COASTAL MODELING AND COASTAL MEASUREMENTS IN THE NETHERLANDS.

by W.T.Bakker 1) and H. de Vroeg. 2)

ABSTRACT.

This paper deals about data, available in the Netherlands for the purpose of coastal modeling. A gap exists between data and coastal models. One reason is "noise" in the data caused by irrelevant features" (from the viewpoint of the model engineer) like breaker bars. Another reason is the structural inaccuracy, especially found in estuary models, caused by the inaccuracy of sand transport formulae and multiple subtraction of big numbers. It is concluded, that coastal modeling can only give reliable answers (in the future) in areas where changes in the boundary conditions are large.

1. INTRODUCTION.

In this paper some information will be given concerning data available in the Netherlands, which can be used for coastal modeling.

Firstly, the need for coastal modeling and the kinds of models available will be considered (ch 2); afterwards, accuracy to be reached by coastal modeling will be discussed (ch.3). Then, it is discussed, which data are needed for coastal modeling and which data are available (ch. 4). Finally, conclusions will be drawn.

The coast of the Netherlands consists of three parts (fig. 1):

- The Wadden coast in the North, where tidal flats at about mean sea level are bordered by Wadden Islands;
- the uninterrupted coast of Holland;
- The delta coast, being the estuary of Rhine, Scheldt and Meuse. In this area the Delta project is near to completion; this project includes damming off the various branches of the estuary and the building of a storm surge barrier in the Eastern Scheldt.



fig. 1 Map of the Netherlands

1) Rijkswaterstaat & Delft University of Technology;

2) Delft University of Technology.

2. NEEDS FOR COASTAL MODELING IN THE NETHERLANDS: KINDS OF MODELS AVAILABLE

Mathematical coastal modeling in the Netherlands is needed for prediction of the coastal behaviour:

- under natural circumstances.

Most of the Dutch coastal defence system consists of dunes. Therefore it is necessary to know how the safety of these dunes will deteriorate in the future in order to reserve certain coastal zone for future coastal defence (no construction of houses is allowed in these areas). A prediction of coastal development for about 25 years should be given.

Furthermore, the effect of enhanced sea level rise by the greenhouse effect of CO₂-emittance should be known (long term trend over 100 years).

- as result of human interference.

This deals as well about future effects of human interference in the past as anticipation on the effects of various alternatives of future human interference.

An example of the first case is the development of the outer delta in the Zeeland area after the finishing of the Delta Project. It is necessary for coastal management to predict those development for purposes of planology.

Examples of the second case are wanted predictions for coastal development after reclaiming coastal areas according various options, after the cutting off of river bends in the estuaries, dredging and dumping of materials in the estuaries (shipping channels) according to various options, etc.

Kinds of models in use in the Netherlands for these purposes are:

- line models

The coast is schematized as one coastline (one-line model) or, in a more detailed way as a "line of the beach" and a "line of the inshore" (two-line model).

In the same way, also multi-line models are used sometimes.

Fig 2e shows the development of a sand supply of Gaussian shape in the course of time as compared with to the development of the German Wadden Island of Sylt, fig.2f shows the same according to the two-line theory ([1], see also Appendix).

Fig 3, from Smit [2], shows a simulation of the development of a cross-section of the coast during 5000 years of sealevel-rise with a multi-line model.

- Field models.

Developments with respect to field models are reported by Boer et al [3], from which fig.4 has been derived.

Furthermore, Stive [4] reports on 2D-vertical models, as CROSTRAN (fig.5).

Dune erosion during storm conditions is calculated with the program DUROS.

* all figures, apart from fig. 1 and 6 appear after the text.

3. ACCURACY OF COASTAL MODELING.

This paper deals about presentation of data, needed for calibration and verification of coastal models. Logically, accuracy of data, compared to accuracy of modeling should be considered. However, accuracy should be seen in relation to relevance and relevance in relation to purpose.

To elucidate the line of thought, let us assume, that a coastal model predicts at a certain site a linear trend. What to do with semi-periodical fluctuations in the measurements, not meant in the philosophy of the model (for instance, breaker bars)? Even if the measuring error is zero, from the viewpoint of the investigator the data will show a disturbing "noise". The problem can be solved, when the reason for the disturbance, or anyway the shape of the disturbance, is exactly known or can be reproduced in a model itself. Mostly however, this is not the case: one shall have to deal with the noise in a statistical way.

The accuracy with which the wanted trend can be determined in a case like this is depicted in fig.6b.

Here it has been assumed, that a sinusoidal disturbance (starting from mean position at time $t=0$) is evaluated as a linear trend, using the least-square method. Thus, a "pseudo-trend" results. (dash-dot line in fig.6a).

On the horizontal scale in fig.6b is the period of observation, expressed in periods of the disturbance. On the vertical scale is the magnitude of the pseudo-trend. One unit of the vertical scale equals the rate of change, found by drawing the tangent to the sine-wave in the zero-position (dashed line in fig. 6a). This is the pseudo-trend found, when the period of observation is very small with respect to the period of the disturbance. It will be called the "reference rate of change". Fig.6b shows, that one needs an observation period of at least 4 to 5 times the period of the disturbance to confine the effect of the disturbance to a small percentage.

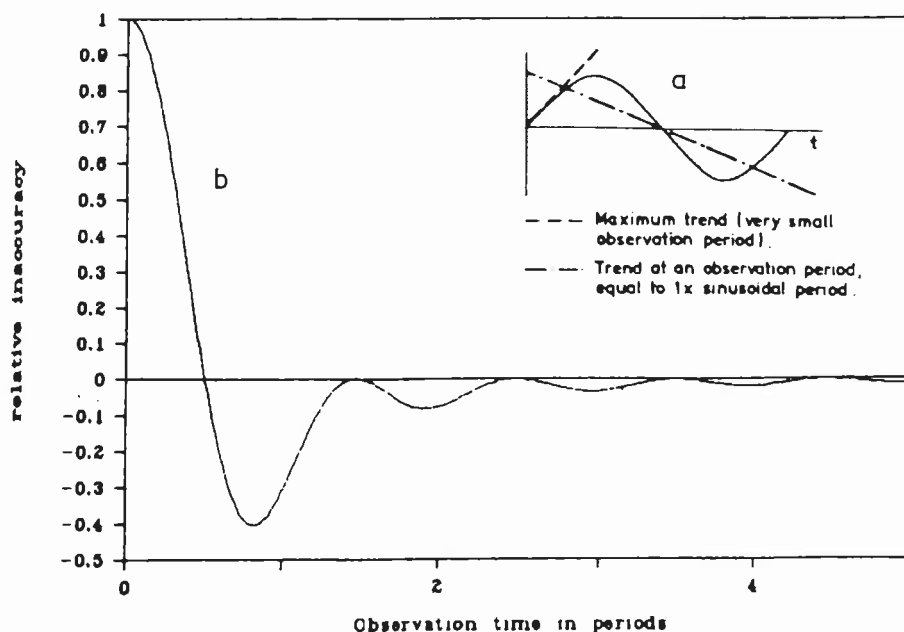


fig. 6
Inaccuracy
trend

For example, along the coast of Holland breaker bars occur, of which the period varies (from place to place, ch.4) between 5 and 20 years, having an amplitude (half of the height) of about 1 m. Thus the reference rate of change is of the order 1.2 m/year, resp. 0.3 m/year. After an observation period of 20 years, the pseudo-trend is of the order of 0.02 m/year, resp. 0.1 m/year (vertical changes). As the mean coastal slope is about 1:100, this means that it is difficult to distinguish the effect of coastal erosion of the order of 2, resp. 10 m/year from this noise. As an illustration, fig.7 shows the linear 20-year trend of the vertical motion of a coastal area North of IJmuiden. The broad zones of accretion and erosion refer to the motion of breaker bars. Southern of IJmuiden (fig7), where the period of the breaker bars is ca. 4 years, the effect on the trend is less distinct.

Another reason, that a nice set of data doesnot include a nice matching on mathematical models is caused by the inaccuracy, inherent of those coastal models. Investigations on the accuracy of coastal models in a tidal estuary like that of the Western Scheldt were carried out by van Rijn [5] and others. Fig. 8 shows this estuary. Accretion of a certain area (one of the sections in fig.8) occurs as a result of a gradient in transport, i.e. the difference in transport in two cross-sections. When each of those sections contains an ebb-channel and a flood-channel, having each as well an ebbtransport and a flood-transport, so many subtractions of big numbers have to be carried out, that a very poor accuracy is the result. Van Rijn finds in a certain area an erosion of 0.3 cm/year with a standard deviation of 35 cm/year [5]. Under those circumstances any resemblance between model and data is purely accidental. In the Netherlands, the only present goal of physical-mathematical coastal modeling is to predict changes at places, where large changes can be expected because of large changes (natural or man-induced) of the boundary conditions.

4.AVAILABLE DATA

Coastal modeling can be either confection, either tailor-made. Confection coastal modeling results from a desk-study, making rules for fashionable behaviour of coasts, after which it is checked, whether a particular coast behaves properly.

Tailor-made coastal modeling starts with a field investigation of a particular coast, followed by a phenomenological description; after this it is investigated, whether there is common sense with respect to continuity and dynamic equations concerning the registered dynamics of the coast.

An elegant example of this kind of modeling is the use of Kalman filtering of series of bottom topography data, from which "coastal constants" occurring in one- and two-line models can be derived [6]. This includes "the assimilation of models in data" instead of "assimilation of data in models".

Fig.9 gives an example of the constants s_1 , s_2 and s_y (see Appendix), derived from time-series of increasing length. On

the horizontal scale is the length of the time series. Data needed for confection coastal modeling is rather confined: initial and boundary conditions and some data for calibration. Data needed for tailor-made modeling is principally unconfined: multi-disciplinary groups of experts are recommended in order to obtain data "not meant in one-man's philosophy". Data available in the Netherlands is as well of the first type (meant for confection models) as of the second kind (tailor-made models).

With respect to the first type: often, for a certain kind of project, for instance for some detail of the Delta project, special measurements have been carried out in order to calibrate an existing model and to find the boundary conditions. It does not seem very useful and would be very time- and paper-consuming as well, to give a catalogue of all those measurements. It may be mentioned, that a database of the sediment transport data of the Western Scheldt has been collected during the mentioned study of van Rijn [5].

On the other hand, often measurements are carried out, following a certain scheme, and resulting in a database (numerically or in forms of maps), but without stating beforehand the way('s) of elaboration.

The following examples can be given.

- a. Waterlevels are measured on a regular base at sites, shown in fig.10
- b. Wave heights are measured on a regular base at the sites, depicted in fig.11. Wave directions are measured at the stations EUT, K13, MPN and AUK.
- c. Bottom topography is periodically measured, resulting in sounding maps. Fig.12 shows the schedule of the mapping; numbers indicate the return period in years, after which measuring is repeated.
- d. Yearly, the bottom topography of the coast is measured, in a coastal stretch extending 200 m landward of the sea slope of the dunes on one hand and 800 m seaward of the dunes on the other hand. Mutual distance of the ranges (being perpendicular to the coastline) is 200 m. Every 5 years, additional measurements are carried out reaching up to 2000 m, at a mutual distance of 1 km.
- e. Furthermore, from time to time geological surveys of the coastal area are commissioned by the Rijkswaterstaat to the Governmental Geological Service.
- f. Historical maps of the tidal outlets and of the Delta region are available.

The following example shows, that considering data first and modeling afterwards may lead to another picture of the coast than starting by modeling.

Fig.14a,b depict the history of the Holland coast in the last 20 years.

Fig.13 shows in two three-dimensional figures the bottom topography of the coast in 1970, for the Northern and Southern area respectively. The landward boundary is about at the dune foot, the seaward boundary extends 800 m more seaward. The

plots have a curved (coast)line as reference line (box in fig.14).

Fig.14a,b show three-dimensional plots, in which the profile in a certain range at successive years are plot next to each other. Seaward motion of the breaker bars is indicated by a direction of the crests in these figures from left upward to right downward.

The successive plots deal about ranges having a mutual distance of 10 km with respect to each other (fig.14, box).

The Northern part (km 10 and 20) is highly influenced by the tidal channels in the outlet between the island Texel and Holland. At the last range shown (116.62, near Hook of Holland), between 1871 and 1974 by a sand supply of 19 mln m³ the coast was extended artificially in seaward direction.

Especially South of IJmuiden, the motion of the bars is more or less regular with a seaward velocity of about 60 m/year. (N.B.: the coast is accreting). The period of the motion is about 4 to 5 year.

In the area, Northern of IJmuiden, the motion is much less and the period is much longer, of the order of 20 years. Fig.9 displays the linear coastal 20-year trend of erosion of accretion in this area (vertical motion), which is much influenced by this bar motion (see ch.3).

Apart from the coastal measurements described before, since about 1850 yearly the site of the line of High Water, the line of Low Water and the dune foot has been measured yearly at places along the coast, having a mutual distance of 1 km.

Referred is to Edelman [7] and Bakker and Joustra [8].

5. CONCLUSIONS.

1. Many measurements on coastal behaviour are available in the Netherlands, ranging from historical maps from centuries ago to detailed measurements of the coastal area during the last 20 years.
2. Still, matching data with coastal models will be a tough job. One reason is, that present coastal models do not properly describe formation and behaviour of breaker bars; another one that accuracy of present sand transport formulae is not sufficient to make look-alikes other than accidental.
3. Semi-statistical approaches with assimilation of some physical knowledge into the data (Kalman-filtering etc.) can be promising for coastal prediction.
4. Independant of the availability of data, coastal modeling can only give reliable answers (in the future) in areas where sudden and large changes occur in the boundary conditions.

6. ACKNOWLEDGEMENT.

The autors are indebted to ir. J.P.v.d.Leye' and ir. H.J.Boot for providing information on measurements.

APPENDIX.
FURTHER EXPLANATION OF SOME FIGURES.

Fig.2 and 9: One- and two-line models.

In one- and two- models ([1],[12]) the coast is schematized by one line (coastline, fig 2a) or two lines (line of the beach and line of the inshore, fig 2b).

Longshore transport S in a one-line model is simulated by the equation:

$$S = S_0 - s \frac{\partial y}{\partial x} \quad (1)$$

where S_0 and s are constants; x,y is a coordinate system where the x -axis is in the mean coastal direction and the y -axis in seaward direction.

For the two-line theory, equation (1) keeps its validity when S_1, S_{01}, s_1 and y_1 , resp. S_2, y_2 replace S, y for the beach, resp. the inshore. In this case furthermore an offshore transport S_y is assumed according to:

$$S_y = s_y (y_1 - y_2 + W) \quad (2)$$

where W denotes the equilibrium distance between beach and inshore and s_y is another constant.

Fig. 2c,d show the initial situation of a coastal supply at S_{y1} [13] and fig 2e, resp 2f the "translation" of contour lines to coastlines, resp. lines of beach and inshore. As well measured as computed (hindcasted) lines are indicated. Accuracies of 10 m over a period of one year with respect to beach behaviour could be reached.

Fig.9 shows results of calculated s_1, s_2 and s_y as found directly from coastal data (by calculating local curvature etc.) valid for the coast of the central part of Holland, using a Kalman filtering technique [6],[14].

Fig.3: multi-line model.

In the multi-line model of Smit [2] eq(2) has been applied for all various layers successively, where constants $s_{y1}, s_{y2}...$ were substituted, decreasing for increasing depth. For the higher layers, constants could be used from former field studies, for the deeper regions the substituted values were guessed. With this kind of model, a transgression of the Dutch coast between 5000 BP and 2000 BP, followed by a regression in connection to sealevel rise could be qualitatively explained.

fig.10 to 14: information on data.

Information on data with respect to waterlevels, wave heights and soundings can be derived from: ir. J.P. v.d.Leye', Rijkswaterstaat, Tidal Waters Division, P.O.Box 20907, 2500EX The Hague.

Exact locations of the measuring points are given in [9] and [10]. Information on the way of gathering the coastal data is given in [11]. A data base on sediment transport is available at Delft Hydraulics, Laboratory De Voorst, attn. ir. L.v.Rijn.

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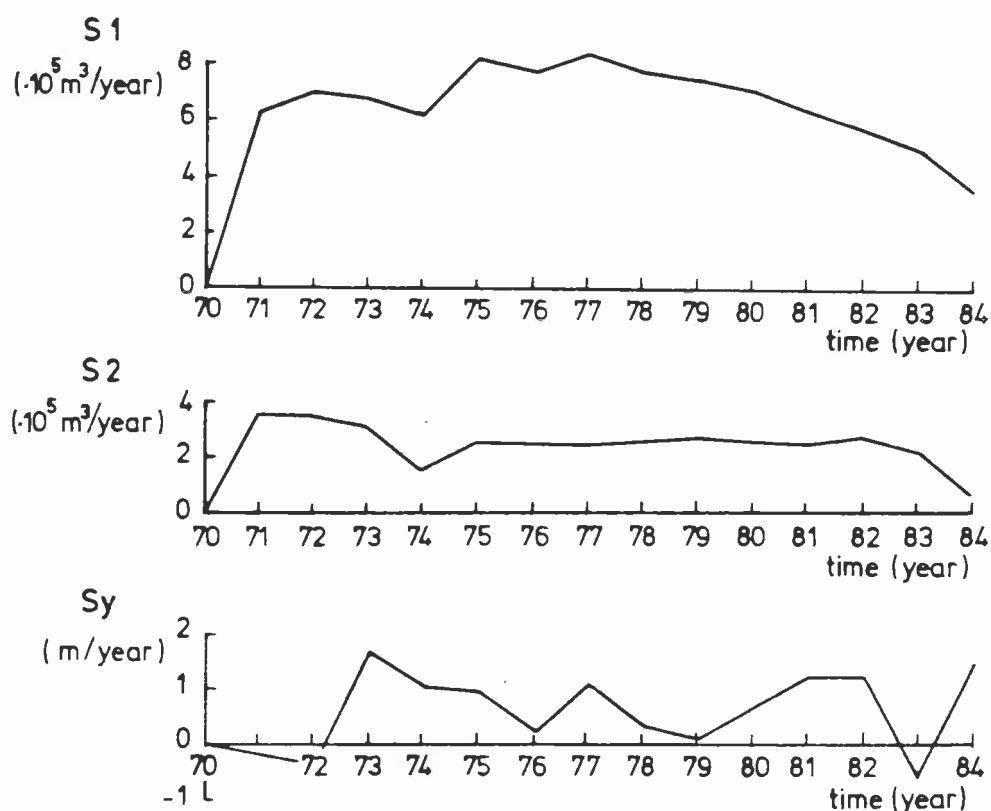


fig. 9. Coastal constants s_1 , s_2 and s_y for central Holland, found by Kalman filtering. After de Vroeg [6].

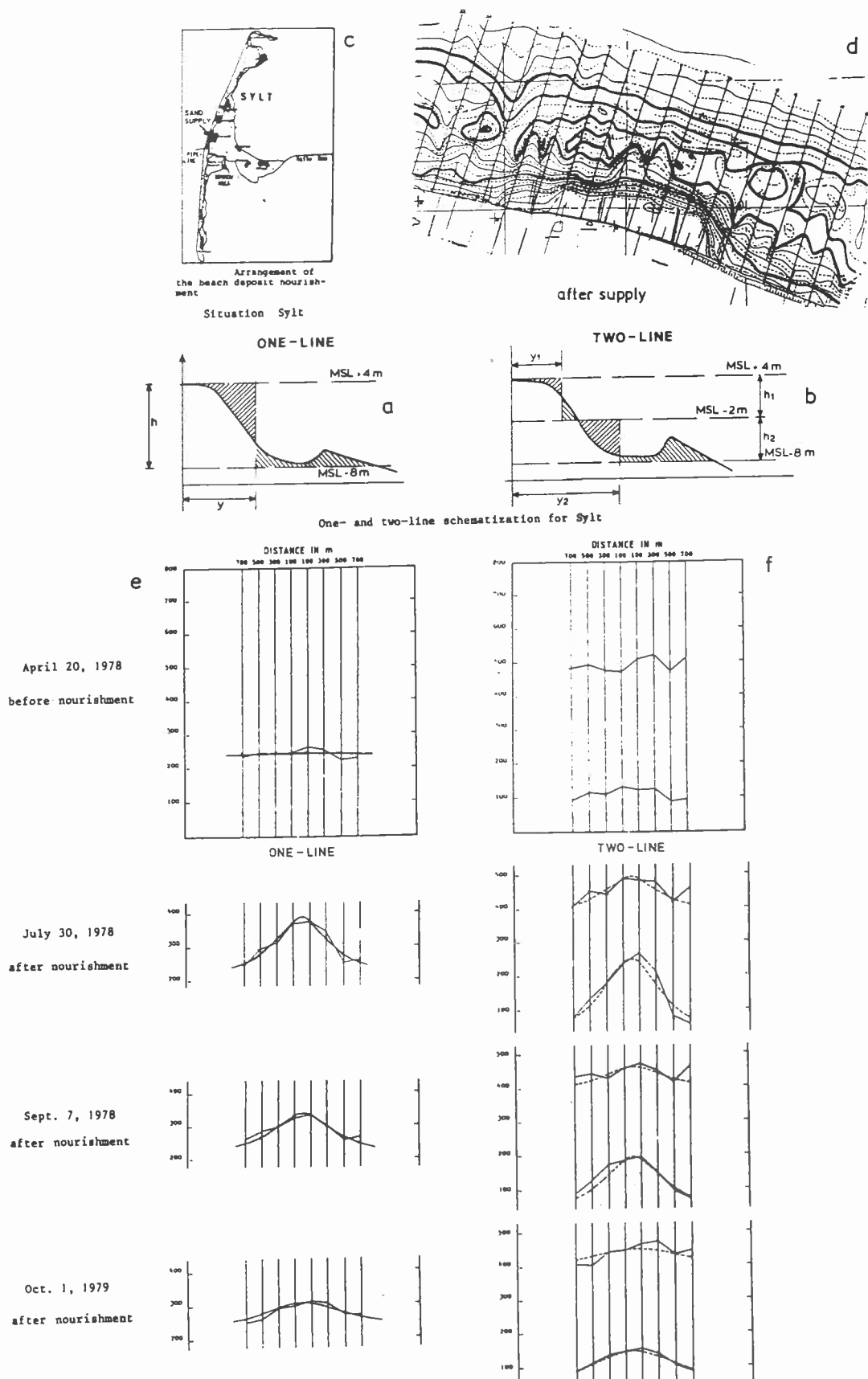


fig. 2. One- and two-line theory, applied on the island of Sylt.

EFFECT OF SEALEVEL RISE (7-LINE MODEL)

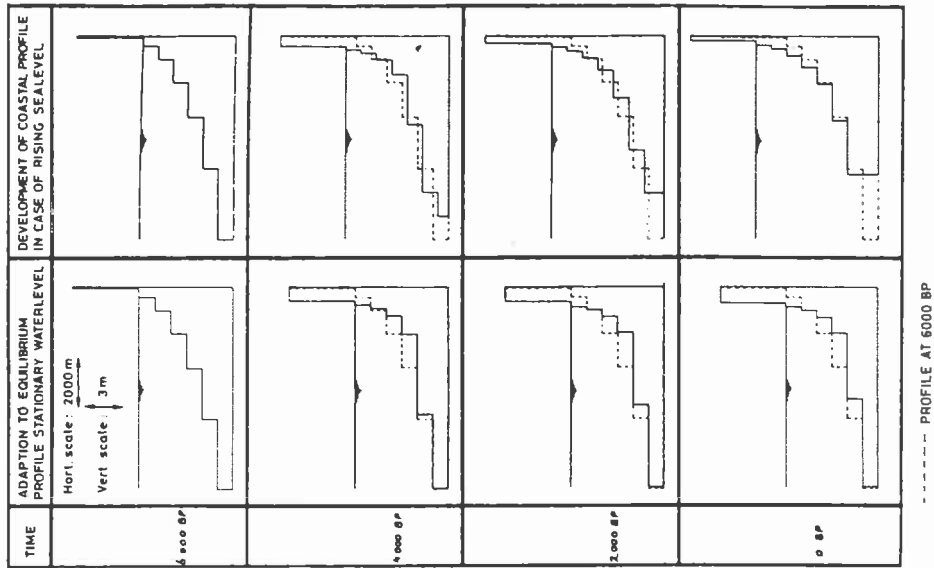
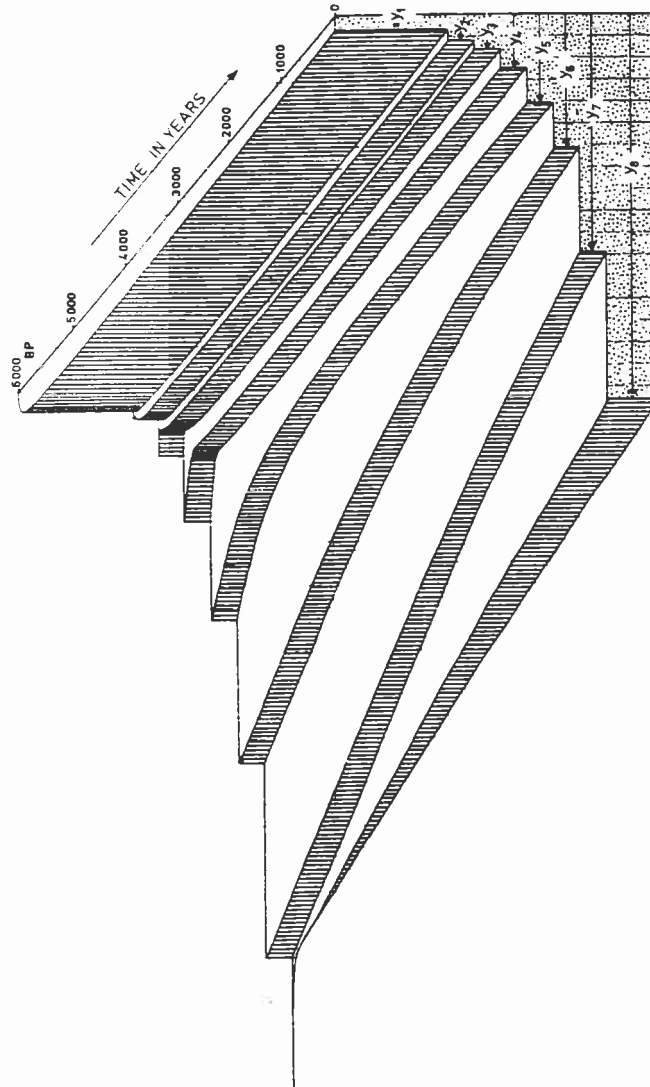


Fig. 3.



3D-presentation of development of profile in case of rising sea level

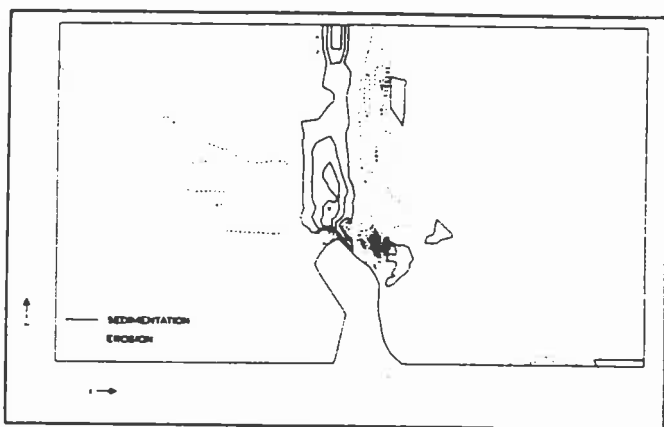
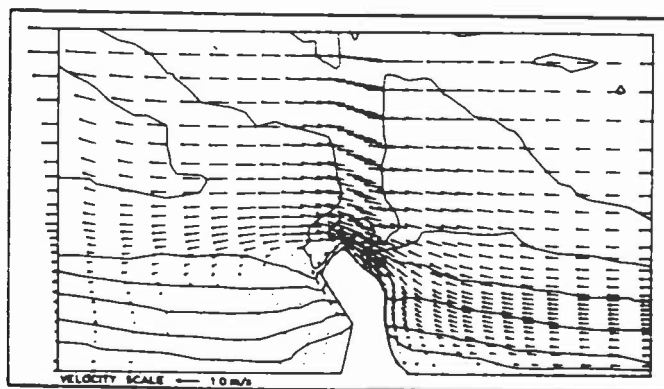
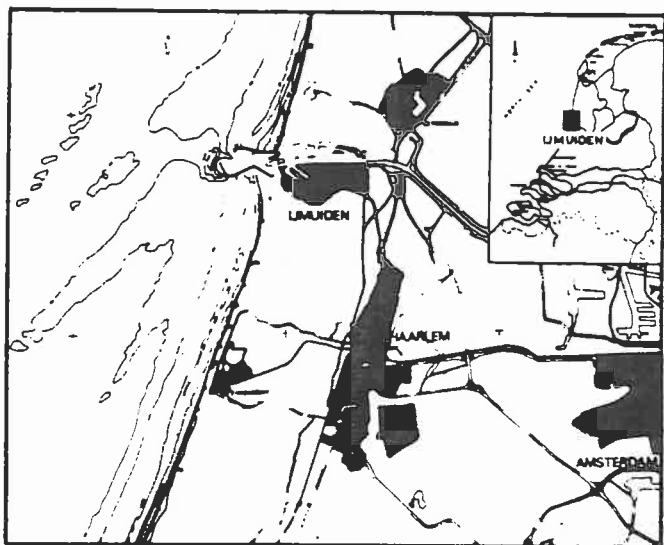


fig. 4. 2D-modeling of flow field and initial sedimentation according to Boer et al. [3].
Vicinity of IJmuiden harbour.

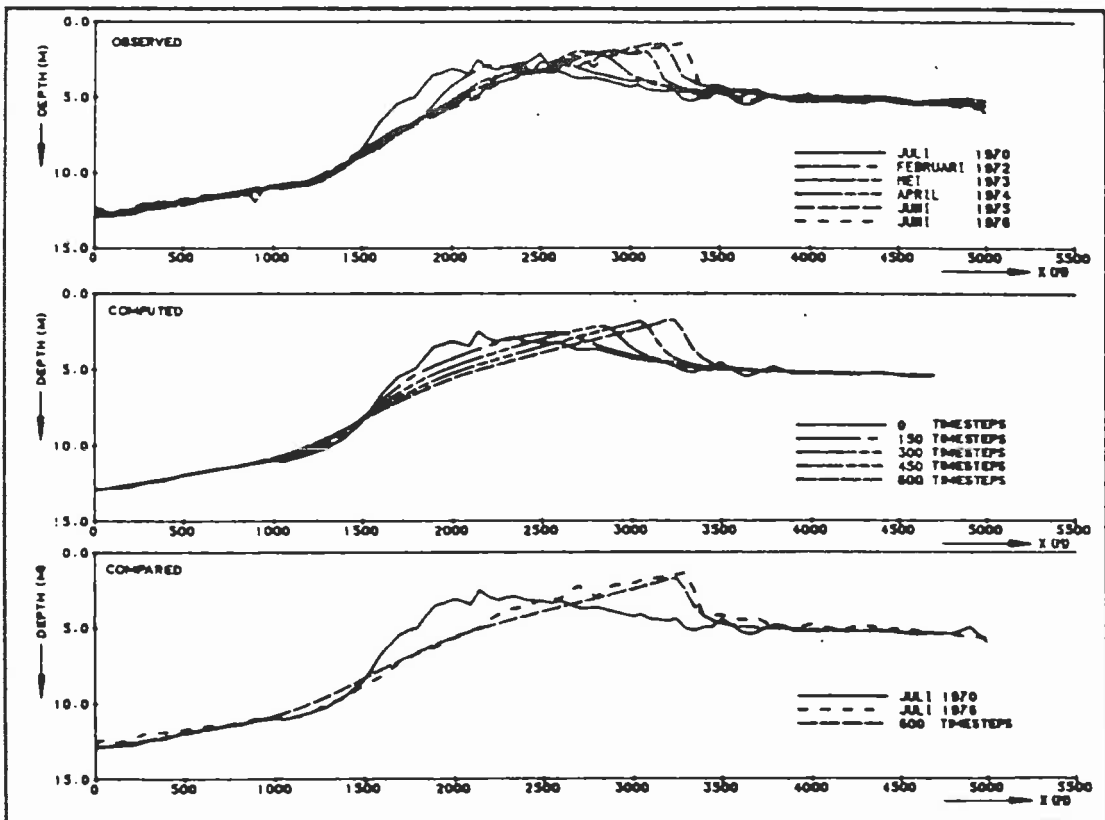
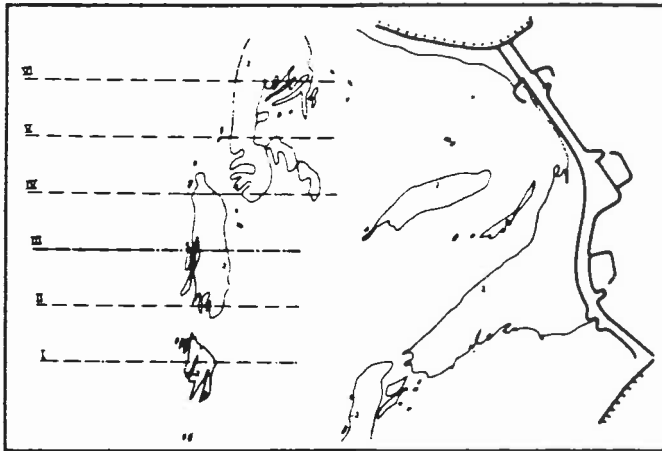


fig. 5. Cross-shore transport modeling according to Stive [4]. Profile evolution of the underwater delta in the former mouth of the Grevelingen estuary
Top: situation in 1976. Bottom: results of model hindcast for section III.

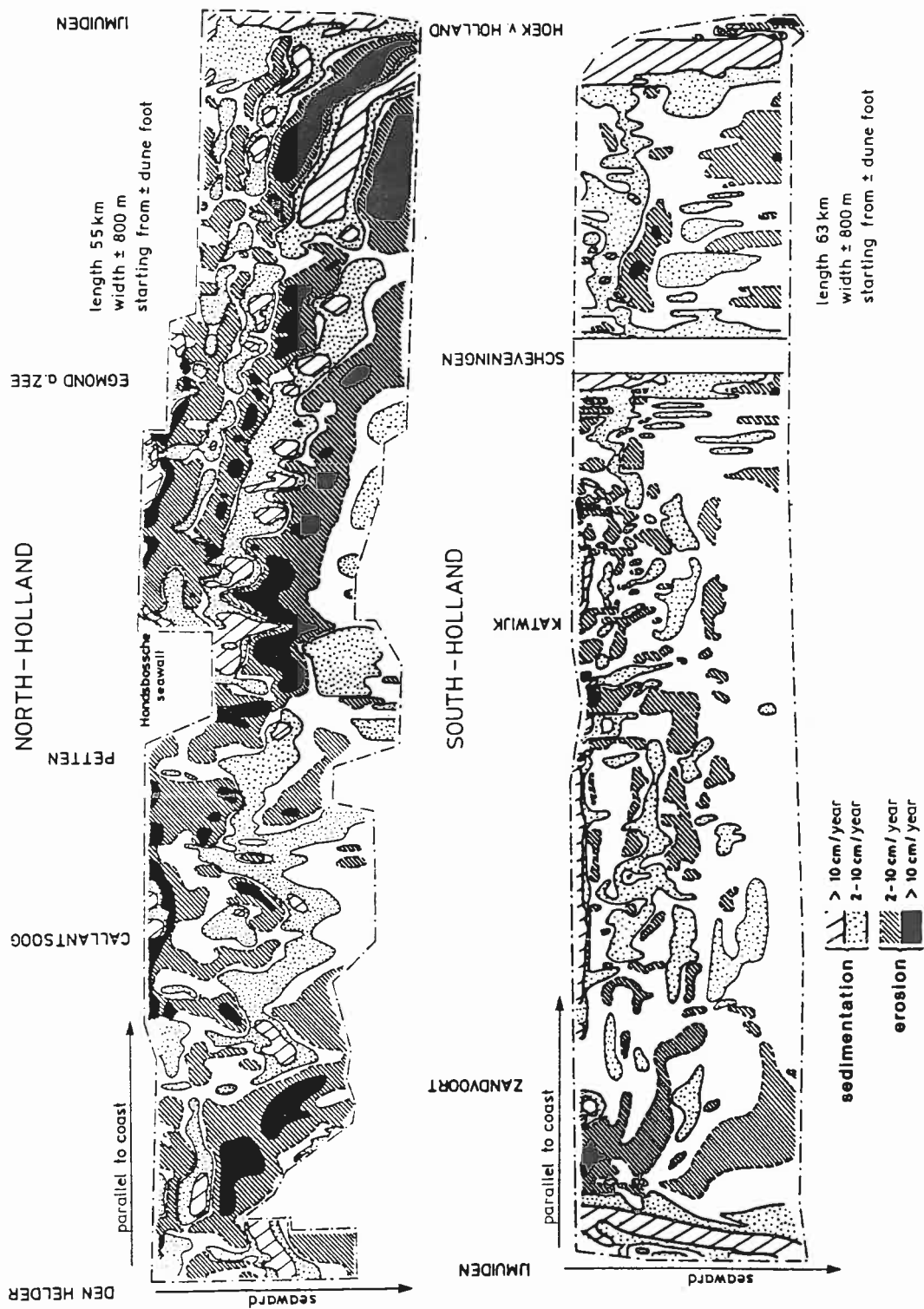


fig. 7. Bottom trends, averaged over 20 years

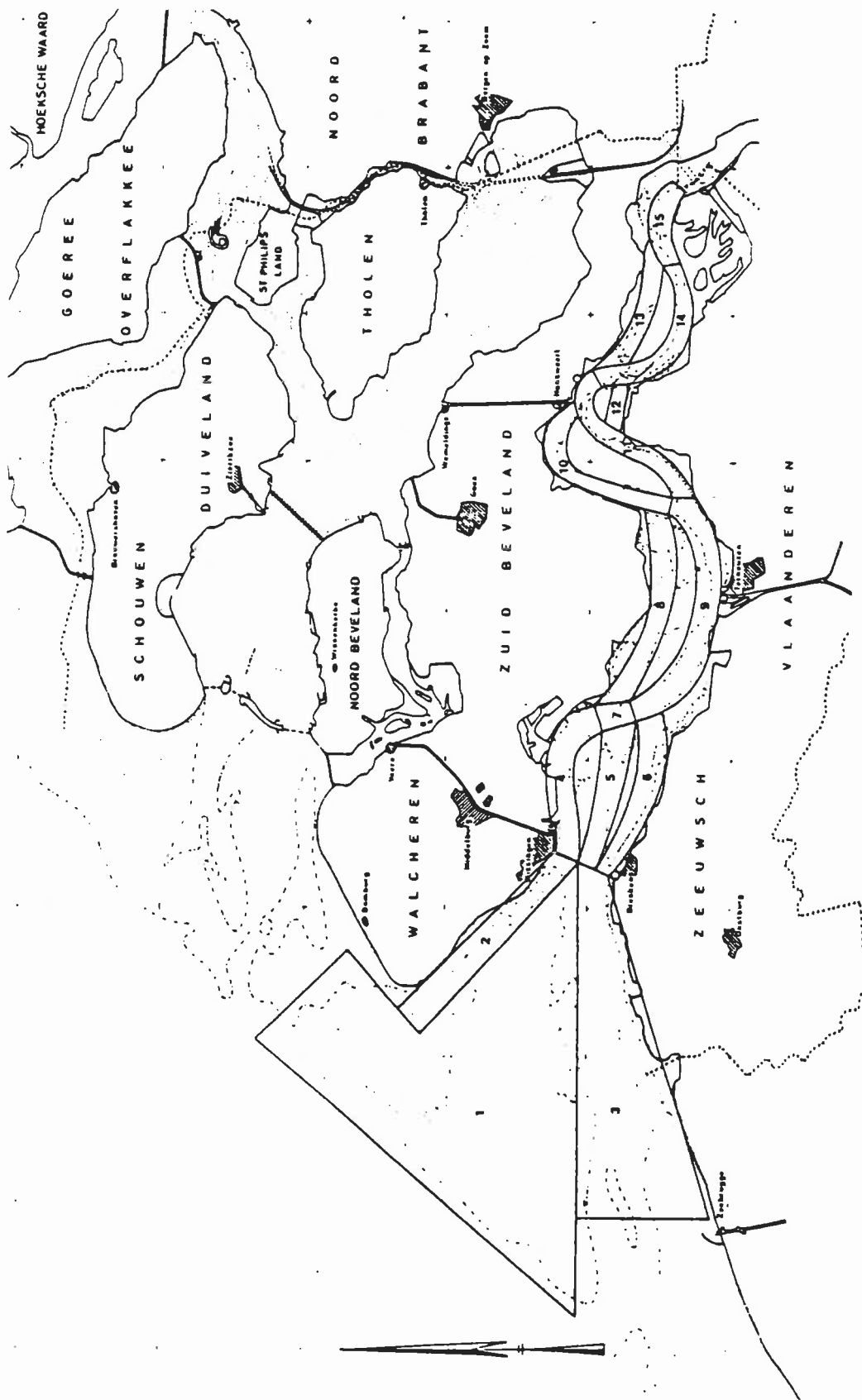


fig. 8. Sections in the Western Scheldt,
in which the transport has been investigated by van Rijn [5].

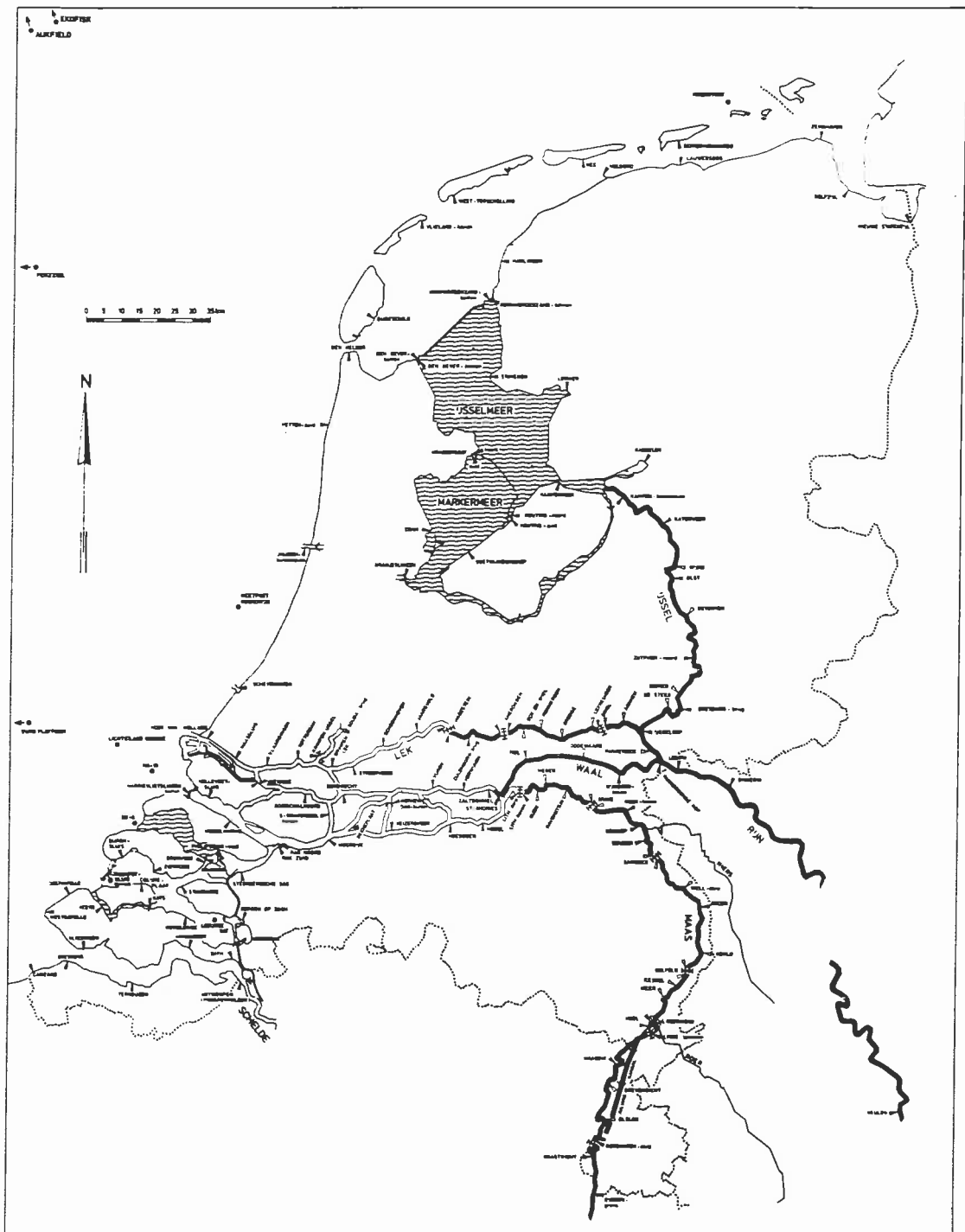


fig. 10. Gauge system for water levels
 • Digital collection; ▽ Recording gauge; ▴ Staff gauge

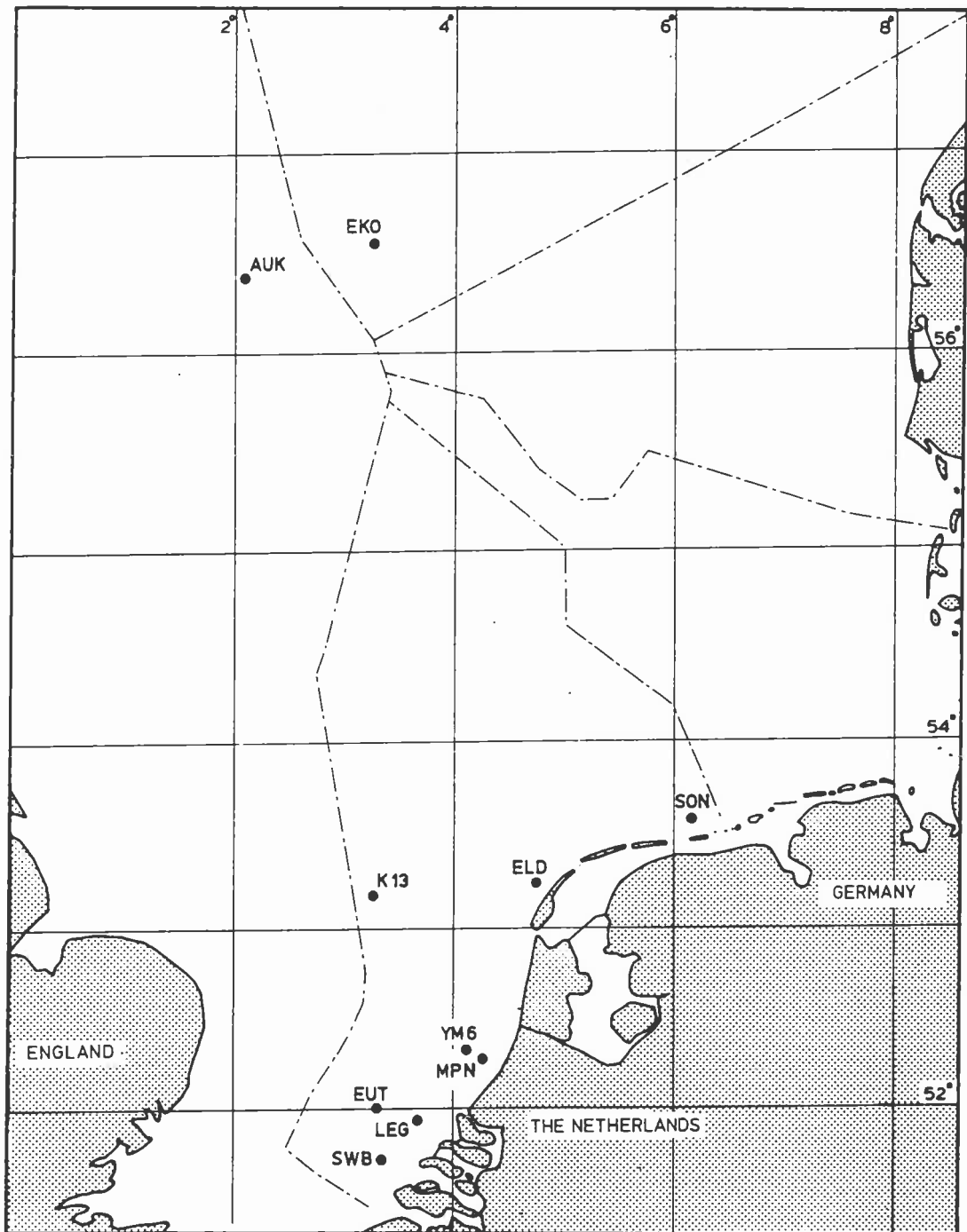


fig. 11. Site of wave gauges.

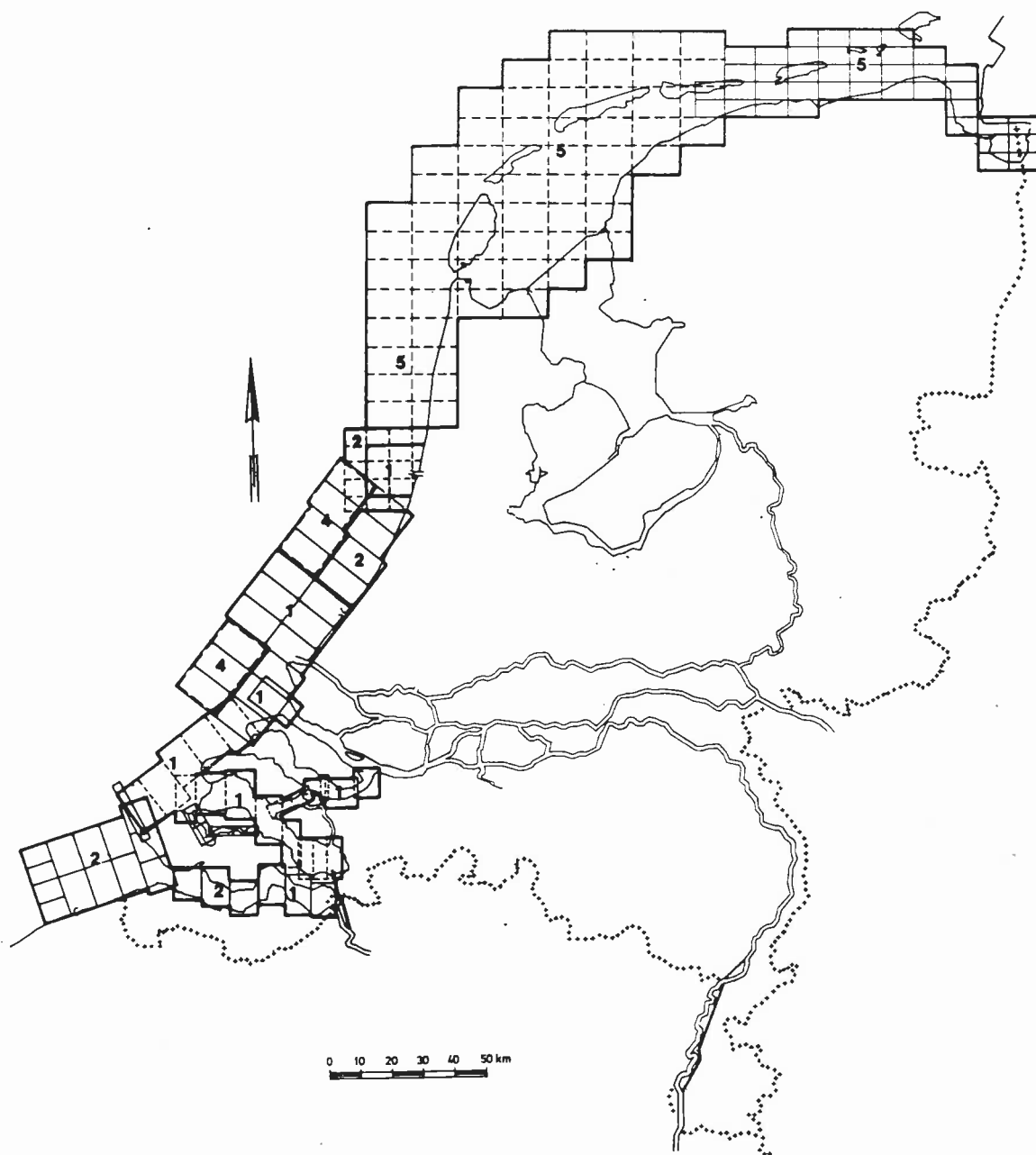


fig. 12. Location maps of underwater topography
The figures indicate the repetition time
of the survey (in years).

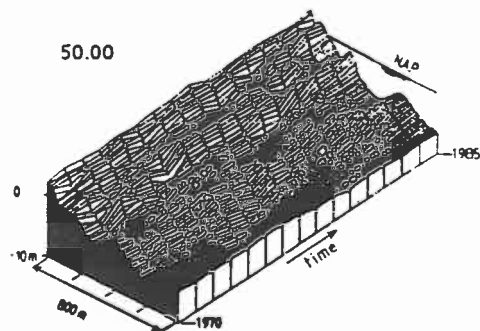
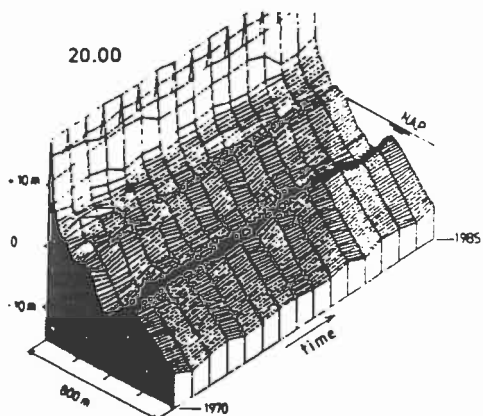
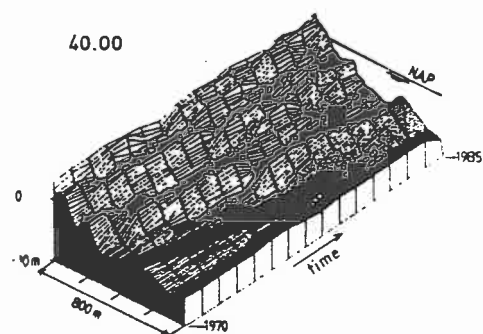
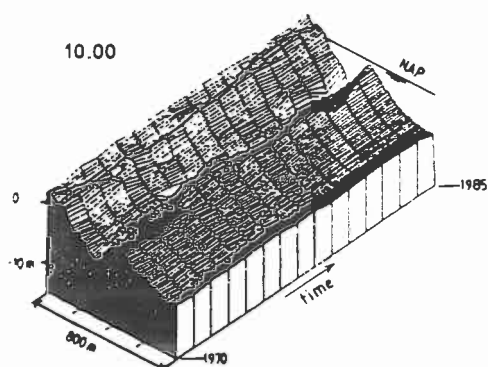
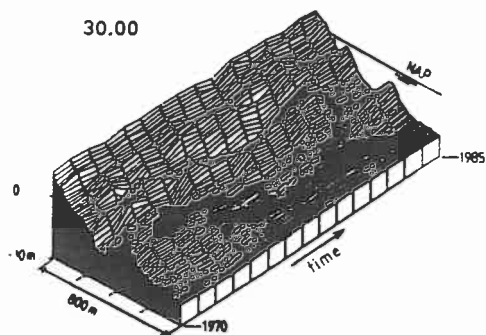
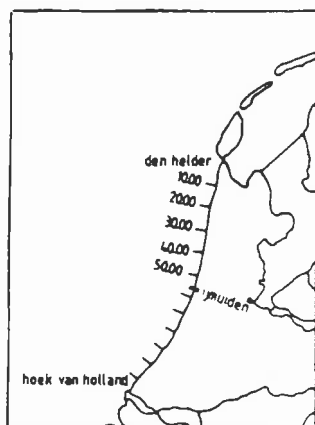


fig. 14a. 3-dimensional time-history diagram for various ranges along the North-Holland coast.

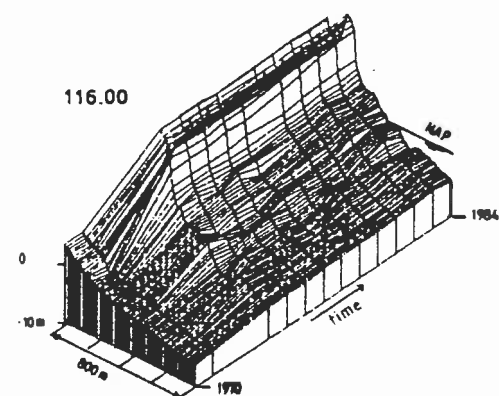
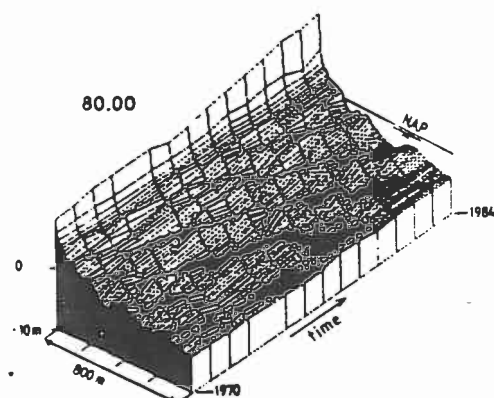
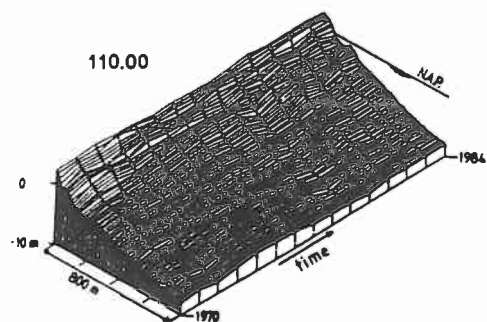
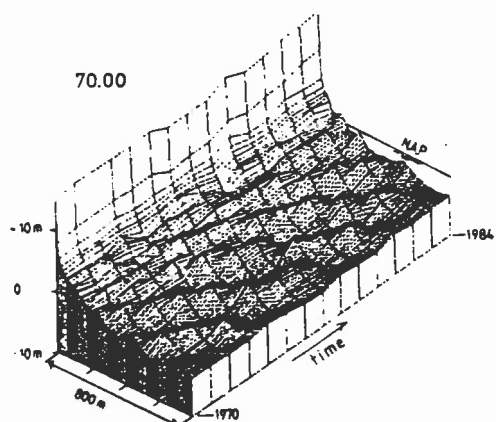
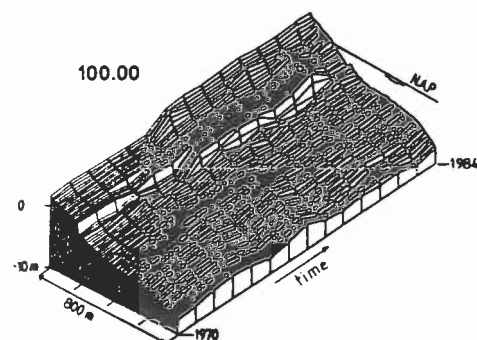
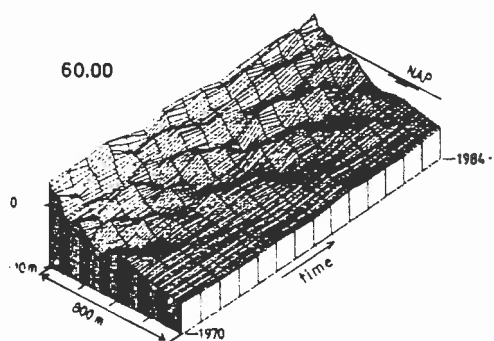
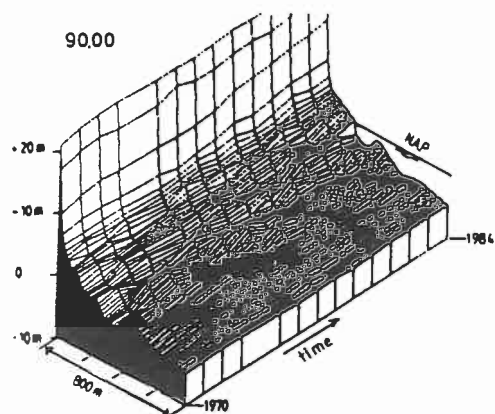
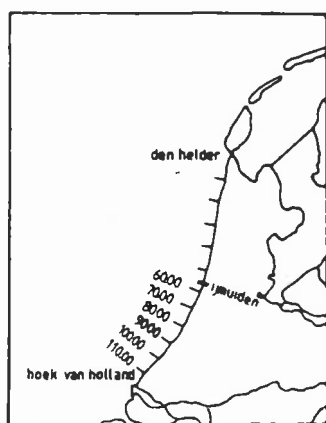


fig. 14b. 3-dimensional time-history diagram for various ranges along the South-Holland coast.